MIL-F-38363B(USAF)
AMENDMENT 3
26 October 1981
SUPERSEDING
AMENDMENT 2
17 October 1979

MILITARY SPECIFICATION
FUEL SYSTEM, AIRCRAFT, GENERAL SPECIFICATION FOR

This amendment forms a part of Military Specification MIL-F-38363B (USAF), dated 13 October 1971, and is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1, under SPECIFICATIONS, Federal: Delete "QQ-P-416, Plating, Cadmium (Electrodeposited)."

PAGE 4

3.2, line 7: Add "Integral tanks when used are part of the fuel system and should comply with material requirements specified for fuel systems."

PAGE 5

3.2.3, delete and substitute:

"3.2.3 Magnesium, copper, and cadmium. Magnesium, copper, and cadmium shall not be used on functional fuel system components that can come into contact with fuel during operation of the aircraft. Cadmium plated fasteners used in areas where contact with fuel can occur shall be overcoated with an Air Force approved fuel tank coating and/or sealant."

PAGE 6

3.5.3, line 7: Add "Static electric charge generation and charge relaxation shall be a major design consideration to preclude the discharge of static electricity in an explosive environment in a fuel tank during all phases of fuel system operation."

3.5.3.1, line 4: Add "The fuel system shall be suitable for use with fuels having resistivity values between 0.1 and 2000 picohms per meter."

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3.5.3.7.1d, delete and substitute:

"d. Inspection criteria for the total plumbing installation including procedures to verify compliance with bonding requirements for fuel tubes, hoses, couplings and clamps."

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PAGE 11

3.5.3.7.3, delete and substitute:

"3.5.3.7.3 Bonding. The aircraft fuel system shall be electrically bonded in accordance with MIL-B-5087. Class S bonding (static charge) is applicable to all fuel system tubing and components, except that there shall be no isolated conducting item permitted inside of fuel tanks or fuel vapor space areas."

PAGE 12

3.5.3.9, delete and substitute:

"3.5.3.9 Lightning and static electricity. The fuel system shall be designed to avoid fire and explosion hazards caused by direct lightning strikes, crossfield streamering or static electricity within the fuel."

PAGE 15

3.7.1.1, line 7 (5th sentence): Delete and substitute "The fuel tank wall shall provide sufficient conductivity to allow relaxing of the electrostatic charge in the fuel."

PAGE 18

3.7.1.6, line 8: Add "A grounding receptacle, in accordance with MS33645, shall be installed for bonding the fuel servicing nozzle to the aircraft structure as specified in 3.11.2.1."

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3.8.2.1, line 10: Add "The generation and dissipation of static electrical charges shall be considered in the voiding design incorporated for each tank. Large voids in the baffle material in areas where high velocity fuel enters the tank are not permitted."

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3.11.1.5, add:

"d. The fuel velocity in any fuel line or component shall be as low as possible to minimize the generation of static electricity in the fuel system: Flow balancing shall be accomplished by line sizing to the maximum extent possible. Orifices used for load balancing shall be located as far upstream from the fuel tank inlet as practical."

3.11.1.6, line 17: Add "Fuel discharge from the level control valve shall be at the bottom of the fuel tank and shall minimize misting, frothing or disturbing the surface of the fuel. Discharge velocitites of less than 10 feet per second are recommended. Multi-discharge outlets should be considered to accomodate total fill rate."

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* 3.12.3 through 3.12.3.3.9, delete and substitute:

"3.12.3 Receptacle/slipway - flying boom. Requirements for the receptacle/slipway - flying boom shall be provided by the contracting activity."

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4.3.3, delete and substitute:

"4.3.3 Aircraft fuel system. The aircraft fuel system used for the ground and flight testing shall be representative of the production aircraft. Prior to the first fueling of flight test or production aircraft, tests shall be conducted to demonstrate compliance with the bonding requirement of 3.5.3.7.3."

NOTE: The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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